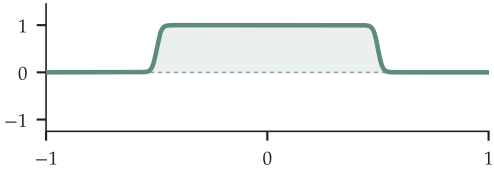
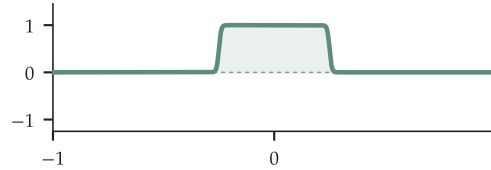
RectBump

Width = 1.0, SlopeWidth = 0.1



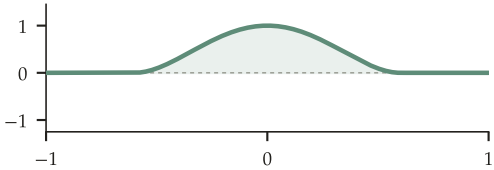
EllipseBump

Width = 0.5, SlopeWidth = 0.1



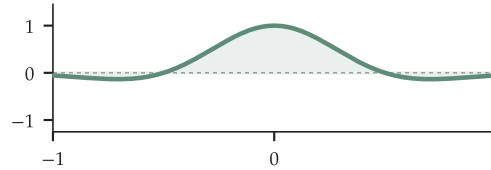
SmoothConvex

$L = 0.6$; exactly Base outside



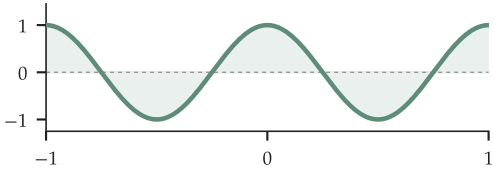
MexicanHat

$L = 0.5$, $\beta = 1$; goes below Base



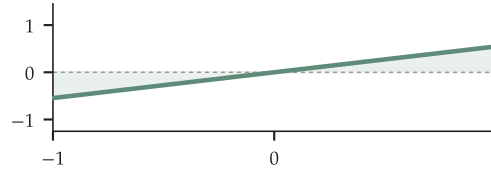
PeriodicWave

Wavelength = 1.0



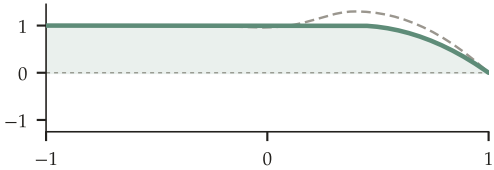
FlatPlane

Angle = 0.5 rad; Base is unused



EdgeBump

Base = Height = 1 (dashed: Height = 1.3), pinned at x_{max}



Five of the seven are
Amp × $f(x)$ + Base.
FlatPlane adds Amp to a tilted plane and ignores Base; EdgeBump holds Base inside and falls from Height to zero at the pinned edge.

Drawn at Amp = 1, Base = 0, Center = 0, EdgeBump at $L_{in} = 0.3$, $L_{out} = 0.6$.

The axis is arbitrary here. In an input file the horizontal parameters are in meters, in the Grid.Xmin to Grid.Xmax coordinates, except EllipseBump.SlopeWidth, which is in normalized radius.